

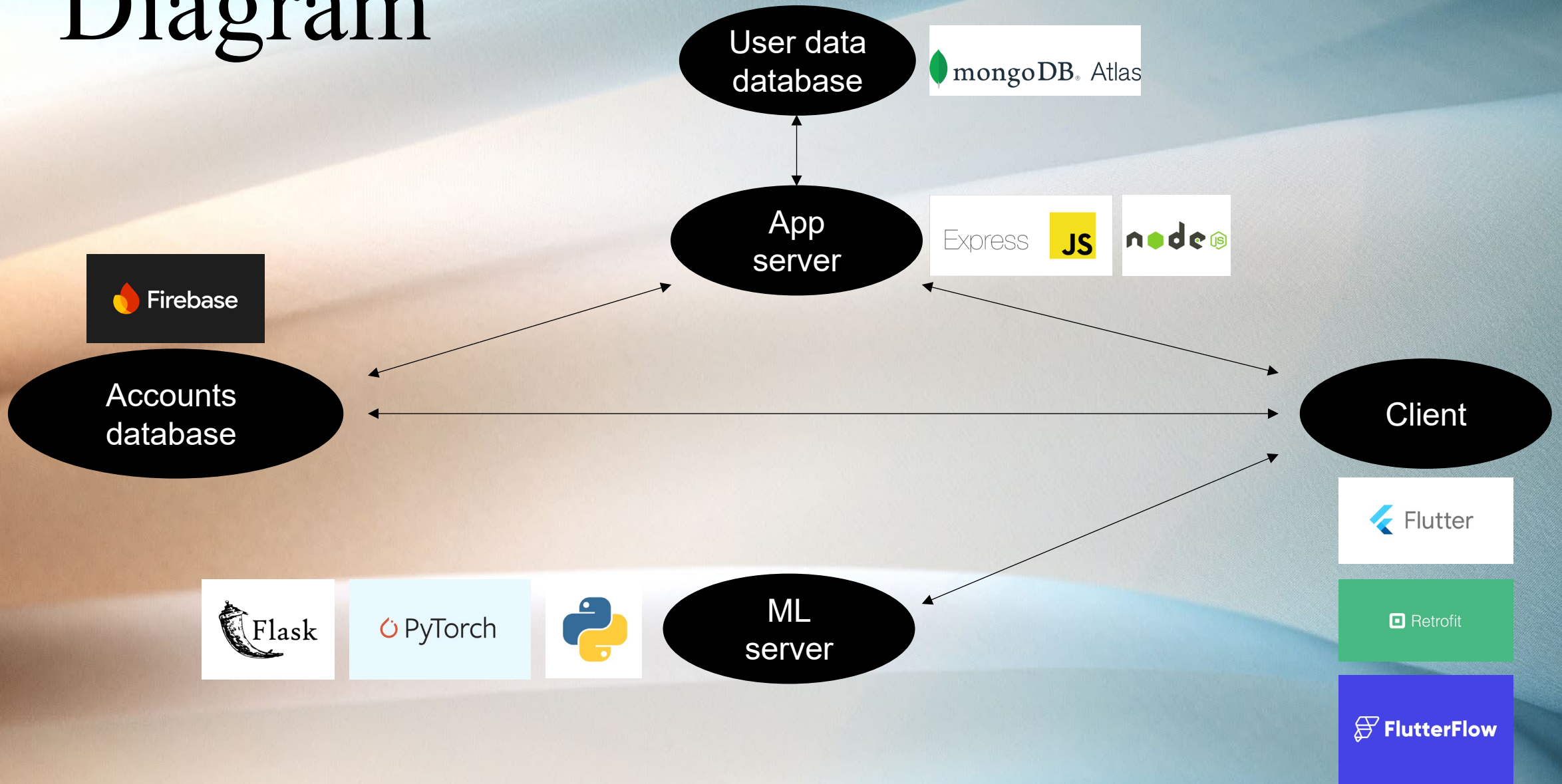


IO - FIT

Project architecture

Ori Arad
Itamar Cohen

Diagram



Tools explanation

Firebase: A platform by Google that provides backend services such as authentication, real-time databases, and cloud functions. It is used here for managing the accounts database and user authentication.

MongoDB Atlas: A cloud-hosted database service for MongoDB. It stores user-specific data efficiently and integrates with the app server.

Node.js: A JavaScript runtime environment that allows running server-side code. It is used in combination with Express.js to handle the app server's backend logic.

Express.js: A minimal and flexible web application framework for Node.js. It is used to build APIs and manage the communication between the client, the accounts database, and the user data database.

Flask: A lightweight Python web framework. It is used here to serve the ML server and expose machine learning functionalities.

PyTorch: A machine learning framework in Python, popular for building and training models. It powers the ML functionalities in the server.

Flutter: A UI toolkit by Google for building cross-platform applications from a single codebase. It is used for creating the client application.

Retrofit: A type-safe HTTP client for Android and Java. It is used in the client to make network requests and communicate with the app server.

FlutterFlow: A visual development platform for creating Flutter apps without deep coding knowledge. It is used to simplify the client development process.

References

ChatGPT: <https://chatgpt.com/>

ML dataset: <https://github.com/siddharthkhincha/Gym-Equipment-Classifier/tree/master>

Exercises dataset: <https://www.bodybuilding.com/exercises> (Sadly the website has been shutdown: **click here for additional information** and **here**)

FlutterFlow: <https://www.flutterflow.io/>